NEPortfolio

UI and UX Design Report | Mobile Application Development | SWE 3409

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Registration number	25/27341
Verification code	MOB-A2-7341
Application	NEPortfolio | Android portfolio application

1. Identity and concept

My name is NZAYISENGA Erneste.

I designed NEPortfolio as a small mobile portfolio that can introduce me to a recruiter or internship supervisor. It presents my education, skills, student projects and professional background in one application.

My target users are recruiters, lecturers and people interested in my work. The key information is available without an internet connection. A user can open the app, read my introduction and navigate between sections within a short time.

Visual direction

I selected a dark navy background, cyan highlights and light text to keep the screens consistent. My illustrated profile picture gives the application a personal identity. Large tap targets and clear labels support easier navigation.



Figure 1. Current Home screen showing my introduction, profile image and verification code.

2. User flow and information architecture

The application uses five bottom navigation tabs. Project Details opens from Projects as an additional stack screen. The Back action returns the user to the project list without losing the current session.

Main area	Action and destination
Home	Read introduction, biography, primary skill and availability
Skills	Browse real skills and education timeline
Projects	View three project names, statuses and summary details
Project Details	Open from a project; read problem, contribution and technologies; return to Projects
Edit Profile	Change headline, biography, primary skill, availability and image; save locally
About	View app information, verification code and handwritten-card photo preview

Main navigation flow

Open app > Home > Select a bottom tab > Read content or make changes

Projects > View details > Project Details > Android Back > Projects

Edit Profile > Enter values / select picture > Validate > Save > Home displays the updated profile

Saved state and media flow

The editor stores the four text fields and the selected local image reference using AsyncStorage. After saving, the app should restore them when opened again. Removing the image returns to the default graphic avatar after saving.

The image selection flow offers Camera, Gallery and Remove. If a user cancels selection or denies permission, the app should give feedback and remain usable.

Offline and online behavior

The core text, project descriptions and saved profile are available offline. External websites, including the project website link, need an internet connection.

3. Screen layout and annotations

These screenshots are from the Android application. They represent the current implementation and are included to show the interface used for the wireframe decisions.



Figure 2. Skills and Education



Figure 3. My Projects

Skills and Education screen

Skills appear as a numbered vertical list with generous spacing. Microsoft Office is kept as one entry covering Word, Excel and PowerPoint. An education timeline follows below the skills list, so the content scrolls naturally on a small phone.

Projects screen

The project list uses small icons, short project descriptions and a clearly visible View details button. Crop Yield Prediction App is labeled In Development to avoid presenting unfinished work as complete.

Each project is expected to open an individual details page with a problem statement, my contribution and the technologies used. These details must remain readable offline.

4. Profile editing and About screen

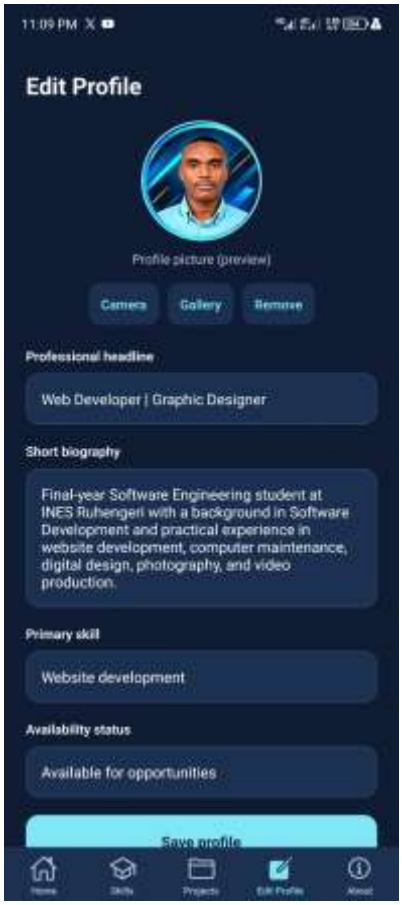


Figure 4. Edit Profile



Figure 5. About NEPortfolio

Profile editor wireframe decisions

The profile image preview appears above three image actions: Camera, Gallery and Remove. Four fields follow: Professional headline, Short biography, Primary skill and Availability status. The Save profile button is placed after the fields to follow the expected form order.

The editor must show a message near an invalid field and prevent saving invalid values. Examples include a headline that is too short, a biography under the required length, and an empty availability field.

About and verification

The About screen displays my full name, registration number, code MOB-A2-7341, program, application version and local-data statement. The handwritten-card preview is used as evidence during the assignment demonstration.

Project Details is a separate stack screen; a full screenshot of that screen should be added after final APK testing if its appearance changes.

5. Visual and accessibility decisions

Color and typography

Element	Design decision
Background	Dark navy (#101D35) for all main screens
Cards	Muted blue (#1D3150) separates content
Highlights and buttons	Bright cyan (#81E6F6) for headings, actions and active navigation
Body text	Light text on dark backgrounds for readable contrast
Typography	Readable sans-serif with clear heading sizes and paragraph spacing

Responsive layout and interaction

I used vertical scrolling and Flexbox-based layouts for narrow Android displays. Content cards expand to the screen width with side padding. Navigation stays available at the bottom, while longer text can wrap instead of being cut off.

I kept buttons and icons large enough to tap and added visible text beside key actions. I also planned labels for image controls, form inputs and navigation. The final APK should be checked with increased system font size to confirm that text remains readable.

Important states to test

Invalid input: show a field-level error without saving. Cancelled gallery selection: remain on the editor. Permission denied: display feedback. No custom image: show the built-in avatar. Offline mode: portfolio screens remain available.

Implementation and review note

Figures 1 to 5 are screenshots from the current app. This report documents my design decisions and the present interface; it does not by itself prove that every function passed physical-device testing. I will update any affected description or screenshot if I modify the final APK.

Tools used

React Native, Expo, TypeScript, React Navigation, Expo ImagePicker and AsyncStorage. Visual report prepared in Word with screenshots from my own app. AI assistance used for drafting and formatting, to be disclosed in AI_USE.md.